

(GR-)Magneto-hydrodynamics
(Accretion)

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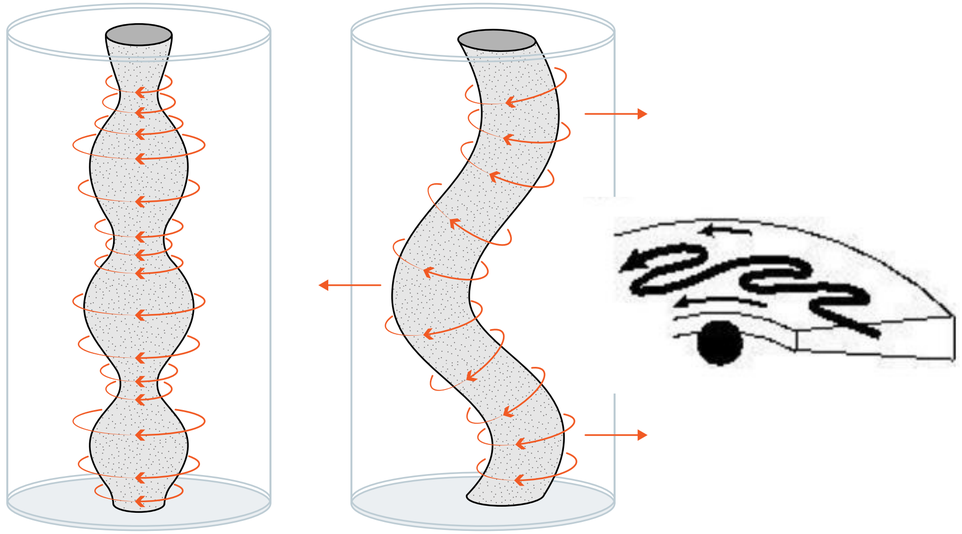
Astrophysical applications of MHD

“Small-scale”

MHD Turbulence
Magnetic dynamos
MHD instabilities
*Reconnection

“Large-scale”

Global accretion flows, disks, winds, jets (planetary, stellar, black holes)
Galactic dynamics, Inter-Stellar Medium,
*Magnetospheres



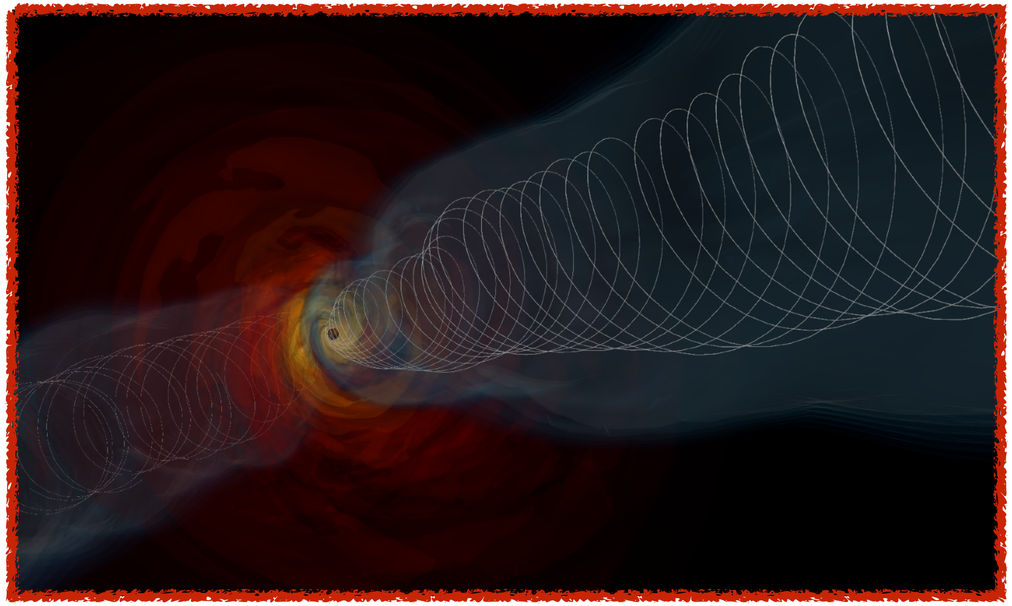
Sausage instability

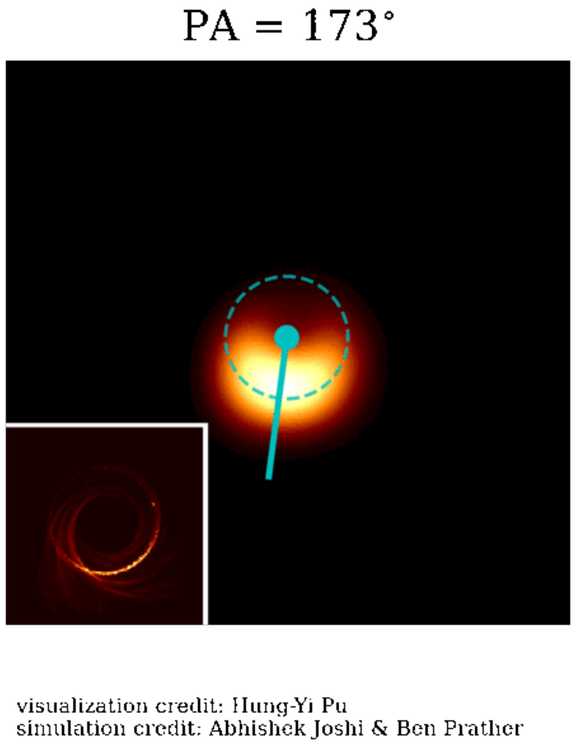
Kink instability



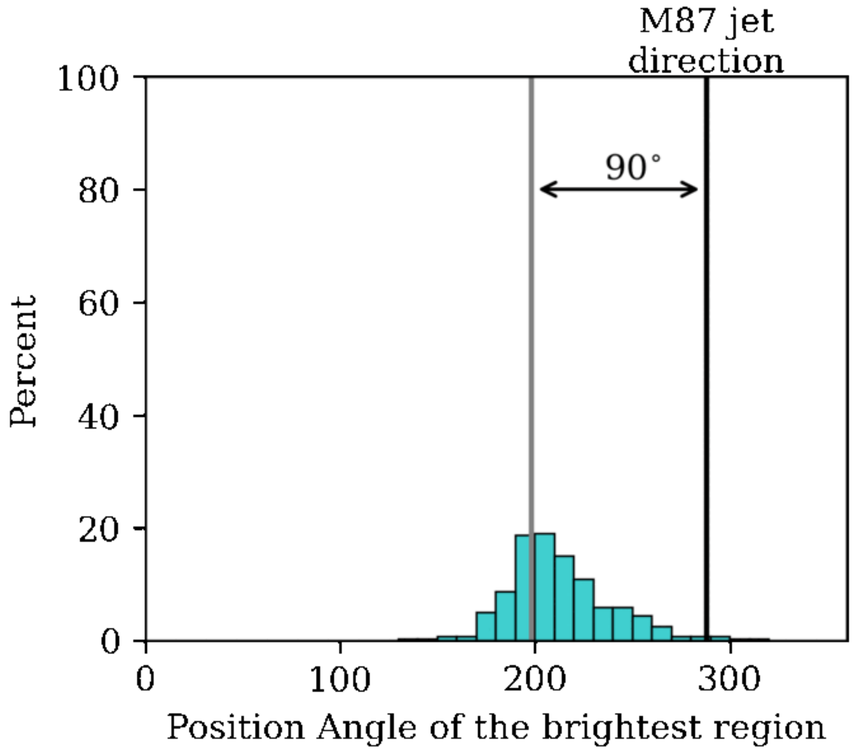
Direct Imaging of Black Holes: Gastronomy

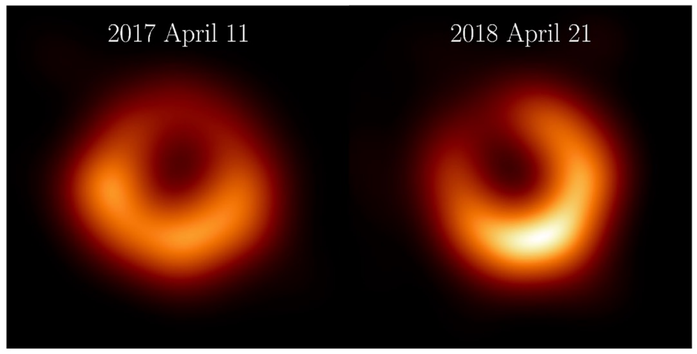
Supermassive Black holes
Messier 87 (M87) & Sagittarius A* (Sgr A*)





visualization credit: Hung-Yi Pu
simulation credit: Abhishek Joshi & Ben Prather





M87 2017, 2018
EHTC+19,+24

Image evolution due to plasma turbulence and gravity

EHTC+25

Magnetohydrodynamics

- Continuum assumption: macroscopic, continuous fluid
- Mean free path of particles, molecules is much smaller than system length scale: collisional (~maxwellian distribution of particles)
- Ideal MHD: isotropic pressure, ignoring viscosity and resistivity
- Quasi-neutrality assumption (any net charges produced get removed on plasma frequency timescales)
- Infinite conductivity in Ideal Ohm's law

$$\overrightarrow{E} + \vec{v} \times \overrightarrow{B} = 0$$

$$\omega_{\mathrm{p}} = \sqrt{\frac{4\pi n_{\mathrm{e}} e^2}{m_{\mathrm{e}}}}$$

- Fluid velocity << speed of light , else relativistic MHD
- Isothermal/adiabatic equations of state

Magnetohydrodynamics

Boltzmann-Vlasov equation in flat space and electromagnetic fields

$$\frac{\partial f}{\partial t} + \vec{v} \cdot \nabla_{\vec{x}} f + \frac{q}{m}\left(\vec{E} + \vec{v} \times \vec{B}\right) \cdot \nabla_{\vec{p}} f = C[f]$$

Collisional term

Taking moments

$$\int (\text{Vlasov equation}) d\boldsymbol{v} \Rightarrow \text{conservation of mass}$$

$$\int (\text{Vlasov equation}) \boldsymbol{v} d\boldsymbol{v} \Rightarrow \text{conservation of momentum}$$

$$\int (\text{Vlasov equation}) \boldsymbol{v}^2/2 d\boldsymbol{v} \Rightarrow \text{conservation of energy}$$

Magnetohydrodynamics Equations

Stress tensor
Advection
Pressure gradient
Magnetic pressure gradient
Magnetic tension (restoring force to fieldline bending)

Continuity equation $$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0$$

Momentum equation $$\frac{\partial (\rho \vec{v})}{\partial t} + \nabla \cdot \left[\rho \vec{v} \vec{v} + \left(p + \frac{B^2}{2\mu_0} \right) \mathbb{I} - \frac{\vec{B}\vec{B}}{\mu_0} \right] = \cancel{0}^{\rho \vec{g}}$$

Energy equation $$\frac{\partial E}{\partial t} + \nabla \cdot \left[\left(E + p + \frac{B^2}{2\mu_0} \right) \vec{v} - \frac{(\vec{v} \cdot \vec{B})}{\mu_0} \vec{B} \right] = \cancel{0}^{\rho \vec{v} \cdot \vec{g}}$$

Advection of energy

Work done by magnetic stresses

where fluid energy is $E = \frac{1}{2}\rho v^2 + \frac{p}{\gamma - 1} + \frac{B^2}{2\mu_0}$
(assumes adiabatic EOS)

Induction equation (using Ideal Ohm's law) $$\frac{\partial \vec{B}}{\partial t} = \nabla \times (\vec{v} \times \vec{B})$$

Gauss Law $$\nabla \cdot \vec{B} = 0$$

There's also entropy conservation $$\frac{Ds}{Dt} = 0$$

Ideal Ohm's Law

Start with momentum equation $m_s n_s \frac{d\vec{v}_s}{dt} = e n_s (\vec{E} + \vec{v}_s \times \vec{B}) - \vec{\nabla} P + \vec{F}_c$

Pressure

Collisions

$s \rightarrow$ ions, electrons

$$v_i = v + \frac{m_e j}{m_i e n} \qquad v_e = v - \frac{j}{en} \qquad v = \frac{m_i v_i + m_e v_e}{m_i + m_e}$$

$$\frac{m_e}{ne^2}\frac{\partial j}{\partial t} = E + v \times B - \frac{1}{ne}(j \times B) - \frac{1}{ne}\nabla P_e - \frac{1}{\sigma} j$$

Collisions

Electron inertia | Lorentz force | Hall effect | Electron pressure

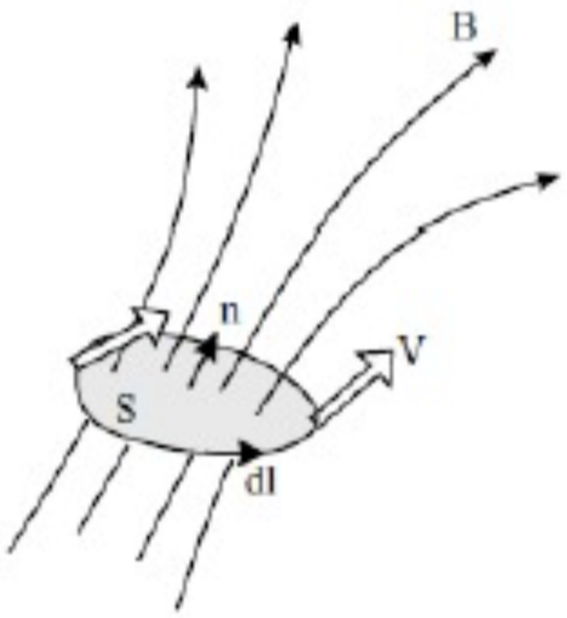
Infinite conductivity plasma $(\sigma \rightarrow \infty)$ and assume fluid changes weakly on typical plasma times and distance scales, therefore we can neglect inertial, Hall and pressure terms for single fluid

Ideal Ohm's Law: $\vec{E} = -\vec{v} \times \vec{B}$

Flux Freezing

Magnetic flux through a surface S, bounded by a curve C

$$\Phi = \int_S \vec{B} \cdot d\vec{S}$$



$$\frac{d\Phi}{dt} = \int_S \frac{dB}{dt} \cdot dS + \int_C B \cdot v \times dl$$

$$\frac{d\Phi}{dt} = -\int_S (\nabla \times E) \cdot dS - \int_S \nabla \times (v \times B) \cdot S$$

$$\text{or,} \frac{d\Phi}{dt} = -\int_S \nabla \times (E + v \times B) \cdot dS = 0$$

Maxwell's law

$$\frac{\partial B}{\partial t} = -\nabla \times E$$

Stokes theorem

Ideal Ohm's Law

Alfvén's theorem

Plasma constrained to move together with magnetic fields

Waves in a homogeneous medium

Monochromatic plane wave propagating through a medium $\psi = \psi_0 e^{i(\vec{k}\cdot\vec{x}-\omega t)} = \psi_0 e^{i\phi}$

Surfaces of constant phase ϕ move in the $\vec{k}$ direction with phase velocity $\vec{V}_{\mathrm{ph}} = (\frac{\partial \vec{x}}{\partial t})_{\phi} = \frac{\omega}{k}\hat{k}$

Wave dispersion relation: spreading of a wave packet $\omega = \Omega(\vec{k})$

e.g., EM waves in vacuum (no charges and no currents): $\nabla\cdot E = 0,\ \nabla\cdot B = 0$

$$\nabla \times E = -\frac{1}{c}\frac{\partial B}{\partial t} \xrightarrow{\text{curl}} \nabla\times(\nabla\times E) = -\frac{1}{c}\frac{\partial}{\partial t}(\nabla\times B)$$

$$\nabla\times B = \frac{1}{c}\frac{\partial E}{\partial t}$$

$$\Rightarrow \nabla(\nabla\cdot E) - \nabla^2 E = -\nabla^2 E = -\frac{1}{c^2}\frac{\partial^2 E}{\partial t^2}$$

Plane wave solution $\vec{E}(\vec{r},t) = \vec{E}_0 e^{i(\vec{k}\cdot\vec{r}-\omega t)} \Rightarrow -k^2 E_0 = -\frac{\omega^2}{c^2}E_0 \Rightarrow \omega = ck$

Linear, non-dispersive

Waves in MHD

How does information spread in MHD flows?

Consider small perturbations in a uniform, static background magnetized plasma

Background equilibrium: velocity $\vec{v}_0 = 0$, magnetic field $\vec{B}_0 = \text{const}$,
plasma density $\rho_0 = \text{const}$, plasma pressure $p_0 = \text{const}$

Adiabatic equation of state: $p \propto \rho^{\Gamma}$ and, $c_s^2 = \partial p / \partial \rho$

Linear perturbations: $Q = Q_0 + \delta Q, \text{ where, } Q \equiv (\rho, p, \vec{v}, \vec{B})$

$$\omega^2 = k^2 \left[\frac{1}{2}(v_A^2 + c_s^2) \pm \frac{1}{2}\sqrt{(v_A^2 + c_s^2)^2 - 4 v_A^2 c_s^2 \cos^2\theta} \right]$$

Using plane wave solution:

$$\psi \propto e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

Magneto sonic wave speed $c_{\text{ms}} = \sqrt{v_{\text{A}}^2 + c_{\text{s}}^2}$

Alfven speed $v_{\text{A}} = B_0 / \sqrt{\mu_0 \rho_0}$ and, sound speed $c_{\text{s}} = \sqrt{\Gamma p_0 / \rho_0}$

Accretion flows

Compact objects exert gravitational force on nearby gas, causing infall of matter onto the object, a process called accretion

Accretion powers a huge amount of electromagnetic radiation from stellar-mass and super-massive black holes

Lots of open questions:

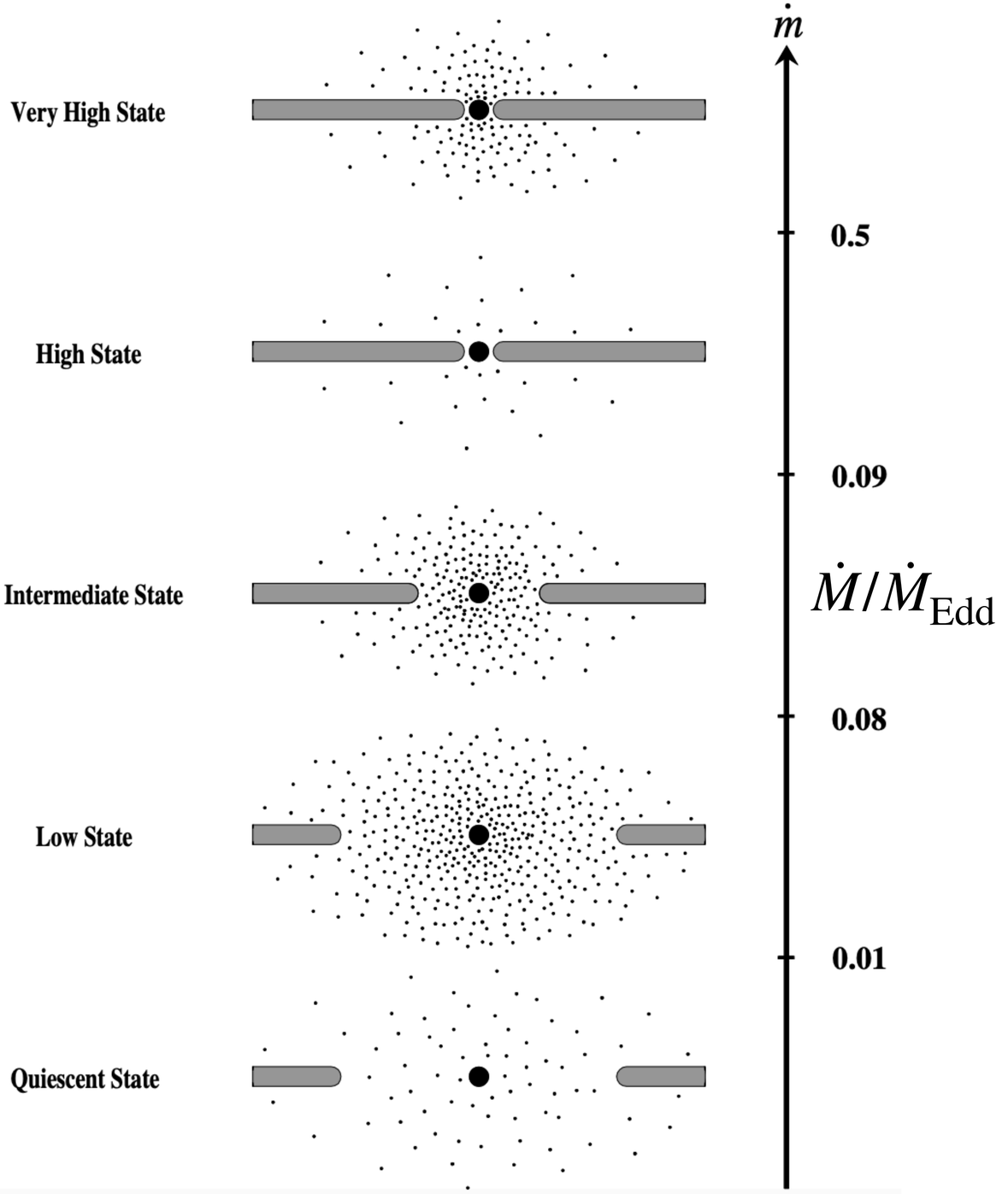
State transitions

Magnetic field creation and transport

Disk instabilities

Angular momentum transport

Radiative properties



Accretion flows (Newtonian, z-integrated, axisymmetric, self-similar)

Mass $$\frac{d}{dR}(\rho R H v) = 0,$$

$$H \sim c_s/\Omega_K$$

Radial mom. $$v\frac{dv}{dR} - \Omega^2 R = -\Omega_K^2 R - \frac{1}{\rho}\frac{d}{dR}(\rho c_s^2),$$

q^+ Energy generated by viscosity

q^- Radiative cooling

Ang. mom. $$v\frac{d(\Omega R^2)}{dR} = \frac{1}{\rho R H}\frac{d}{dR}\left(\nu\rho R^3 H \frac{d\Omega}{dR}\right),$$

Ratio of advected energy to heat generated

Energy $$\rho v T\frac{ds}{dR} = q^+ - q^- = \rho\nu R^2\left(\frac{d\Omega}{dR}\right)^2 - q^- \equiv f\nu\rho R^2\left(\frac{d\Omega}{dR}\right)^2 = q^{\rm adv}$$

$q^+ \simeq q^- \gg q^{\rm adv} \Rightarrow$ Cooling dominated: Thin disk (e.g., Shakura & Sunyaev, Novikov Thorne)
SLE disk (Shapiro, Lightman & Eardley): optically thin, hot flow,
thermally unstable as heating puffs up the disk, reducing ability to cool

$q^{\rm adv} \simeq q^+ \gg q^- \Rightarrow$ Advection-dominated accretion: hot flow where cooling does not play a role

Negligible heat generation occurs in simple flows like Bondi accretion

Narayan+98

Thin disk solution (weakly magnetized)

2D steady flow ($\partial/\partial t = 0 = \partial/\partial\phi$), thin disk ($H \ll R$), Newtonian gravity

$$\Phi(R,z) = -\frac{GM}{\sqrt{R^2+z^2}}$$

Mass $\quad \frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\vec{v}) = 0 \quad \Rightarrow \quad \frac{1}{R}\frac{\partial}{\partial R}(R\rho v_R) + \frac{\partial}{\partial z}(\rho v_z) = 0$

Radial mom. $\quad \rho(\vec{v}\cdot\nabla)\vec{v} = -\nabla P - \rho\nabla\Phi + \frac{1}{4\pi}(\nabla\times\vec{B})\times\vec{B}$ gives $\quad \rho\left(v_R\frac{\partial v_R}{\partial R} - \frac{v_\phi^2}{R}\right) = -\frac{\partial P}{\partial R} - \rho\frac{\partial\Phi}{\partial R}$

Vertical mom. $\quad 0 = -\frac{\partial P}{\partial z} - \rho\frac{\partial\Phi}{\partial z} + \frac{1}{4\pi}\left[(\nabla\times\vec{B})\times\vec{B}\right]_z$ gives $\quad \frac{\partial P}{\partial z} = -\rho\frac{GMz}{R^3}$ as $\quad \frac{\partial\Phi}{\partial z} \approx \frac{GMz}{R^3}$

Taking $P \sim \rho c_s^2$, we get $\rho(z) = \rho_0 \exp\left(-\frac{z^2}{2H^2}\right)$ where $\boxed{H = \frac{c_s}{\Omega_K}}$, $\quad \Omega_K = \sqrt{\frac{GM}{R^3}}$

$v_R \ll v_\phi \sim v_K = R\Omega_K$

Angular momentum transport following viscosity parameter notation

Using vertically integrated angular mom eq., and the definition of the accretion rate, we can find $\quad \frac{|v_R|}{v_\phi} \sim \alpha\left(\frac{H}{R}\right)^2$ where $T^{\rm mag}_{R\phi} = \frac{B_R B_\phi}{4\pi} \sim \alpha P$

Advective inflows

Narayan & Yi 94, Narayan+98

Applying radial self-similarity to (M)HD eq. for accretion and reduce cooling

$$v(R) = -\frac{(5+2\epsilon')}{3\alpha^2} g(\alpha,\epsilon')\alpha v_{\rm ff},$$ Rapid accretion

$$\Omega(R) = \left[\frac{2\epsilon'(5+2\epsilon')}{9\alpha^2} g(\alpha,\epsilon')\right]^{1/2} \frac{v_{\rm ff}}{R},$$ Sub-keplerian rotation

$$c_s^2(R) = \frac{2(5+2\epsilon')}{9\alpha^2} g(\alpha,\epsilon') v_{\rm ff},$$ Hot flow

$$v_{\rm ff} \equiv \left(\frac{GM}{R}\right)^{1/2}, \quad \epsilon' \equiv \frac{\epsilon}{f} = \frac{1}{f}\left(\frac{5/3-\gamma}{\gamma-1}\right), \quad g(\alpha,\epsilon') \equiv \left[1+\frac{18\alpha^2}{(5+2\epsilon')^2}\right]^{1/2} - 1.$$

The disk scale height $H \sim R$ implying a thick disk

Which shows broadly

$$\rho(R) \propto R^{-3/2}$$
$$v_R(R) \propto R^{-1/2}$$
$$c_s^2(R) \propto R^{-1} \propto T$$
$$v_\phi(R) \propto R^{-1/2}$$

Applicable for slow accretors (e.g., Sagittarius A*) and super fast accretors (Super-Eddington)

Introducing relativistic MHD

In Newtonian (non-relativistic) MHD, equations are written using 3-vectors and time derivatives in Euclidean space

For systems with black holes, neutron stars, jets, we need relativistic velocities and compatibility with curved spacetimes.

4D spacetime coordinates: $\vec{x} \rightarrow x^{\mu} \equiv (t, \vec{x})$

3-vectors to 4-vectors: $\vec{v} \rightarrow u^{\mu} \equiv \gamma\,(1, \vec{v}); \quad u^{\mu} u_{\mu} = -1$

Convective to Covariant derivative: $D/Dt \rightarrow \nabla_{\mu}$

Velocity is now bounded by speed of light c

Metric convention (-,+,+,+) and let's switch to natural units, $G = c = 1$

Black Hole spacetime metrics

Vacuum solutions to the Einstein's equations

Schwarzschild metric (in Boyer-Lindquist coordinates)

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2$$

Horizon radius

$$r_{\mathrm{H}} = \frac{2GM}{c^2} = 2r_{\mathrm{g}}$$

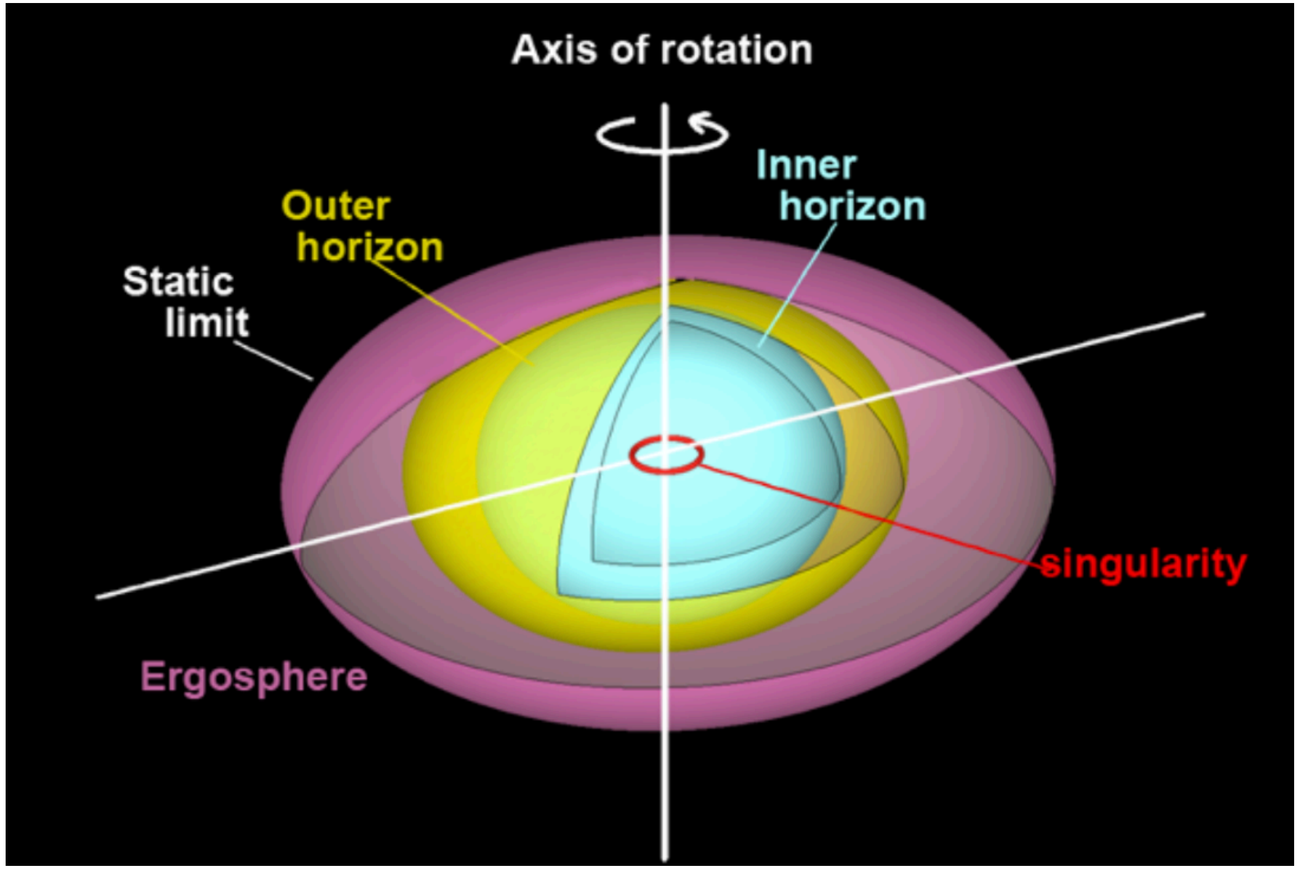
Kerr metric (in BL coordinates)

$$ds^2 = -\left(1 - \frac{2r}{\Sigma}\right) dt^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma\, d\theta^2 + \frac{A \sin^2\theta}{\Sigma} d\phi^2 - \frac{4ar \sin^2\theta}{\Sigma} d\phi\, dt\,,$$

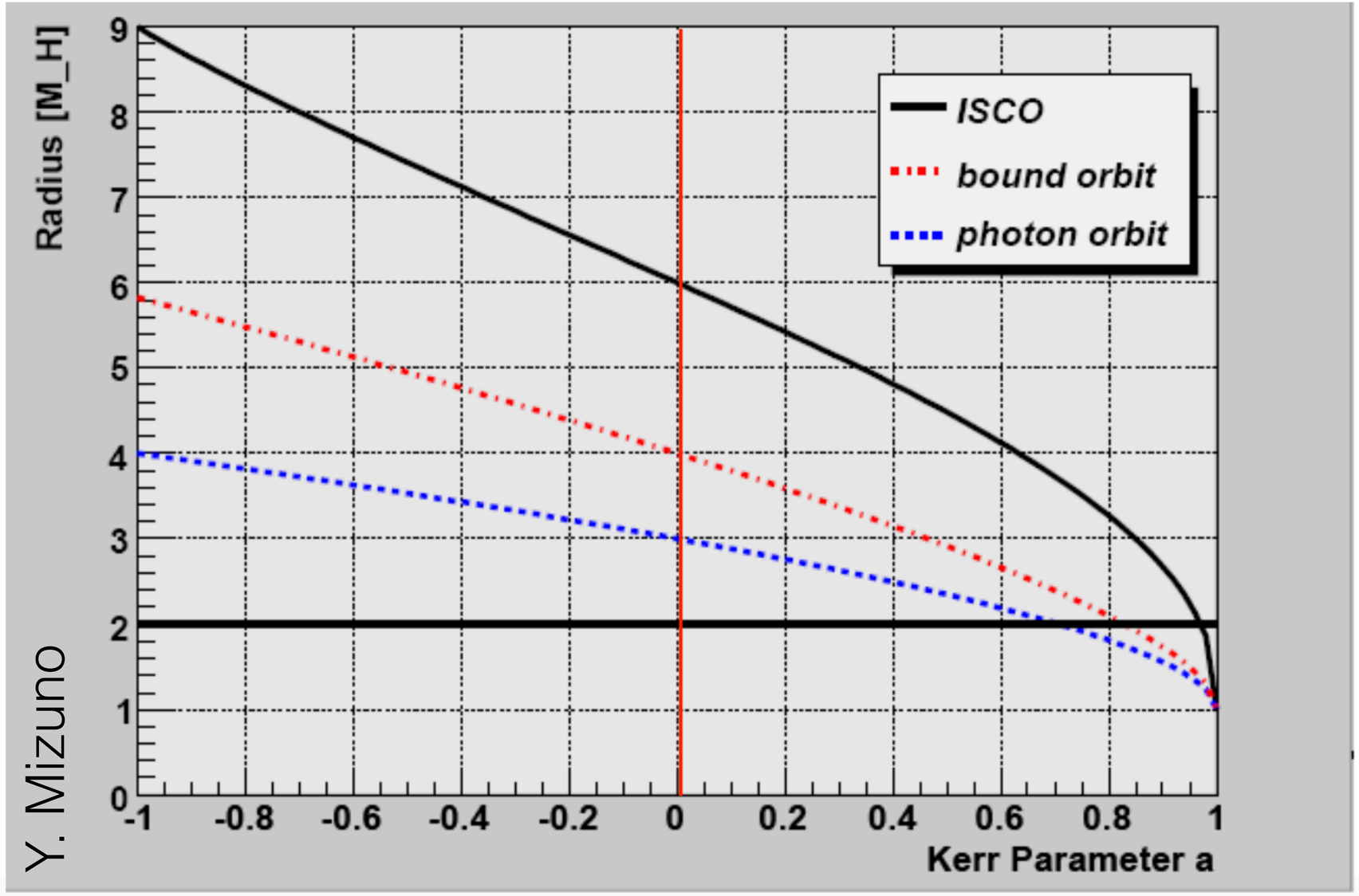
$$\Sigma \equiv r^2 + a^2 \cos^2\theta, \Delta \equiv r^2 - 2r + a^2,$$

$$A \equiv (r^2 + a^2)^2 - a^2 \Delta \sin^2\theta$$

$$r_{\mathrm{H}} = r_{\mathrm{g}}[1 + \sqrt{1 - a^2}]$$



Black Hole spacetime metrics



Plunging streams near the horizon produce lots of radiation e.g., in X-rays

The frame-dragging rate $\Omega = -\frac{g_{t\phi}}{g_{\phi\phi}}$ directly affects the disk evolution

General relativistic MHD formulation

Construct 4-force density $f^{\mu} \equiv dp^{\mu}/d\tau = d(mu^{\mu})/d\tau$

i) spatial components from Lorentz force $\vec{f} = \rho\vec{E} + \vec{J} \times \vec{B}$

ii) time component from temporal derivative of energy $\vec{f} \cdot \vec{u}$

Combining $\Rightarrow f_{\mu} = F_{\mu\nu}J^{\nu}$ where we introduce the Faraday tensor $F^{\mu\nu}$

The Faraday tensor is constructed using the Maxwell equations

$$\nabla \cdot \vec{E} = \rho \;\&\; \nabla \times \vec{B} - \partial\vec{E}/\partial t = \vec{J} \Rightarrow \partial_{\mu}F^{\mu\nu} = J^{\nu}$$

$$\nabla \cdot \vec{B} = 0 \;\&\; \nabla \times \vec{E} - \partial\vec{B}/\partial t = 0 \Rightarrow \partial_{\mu}\,{}^{*}F^{\mu\nu} = 0 \text{ or } \partial_{[\mu}F_{\nu\lambda]} = 0$$

Where we also have the dual of the Faraday tensor ${}^{*}F^{\mu\nu} = (1/2)\epsilon^{\mu\nu\kappa\lambda}F_{\kappa\lambda}$

General relativistic MHD formulation

Define the magnetic 4-vector potential $A^\mu \equiv (\phi, \overrightarrow{A})$
using the scalar potential ϕ and the vector potential $\overrightarrow{A}$

We have our definitions from $\nabla \cdot B = 0 \rightarrow \overrightarrow{B} = \nabla \times \overrightarrow{A}$
and $\nabla \times \overrightarrow{E} - \partial \overrightarrow{B} / \partial t = 0 \rightarrow \overrightarrow{E} = -\nabla \phi - \partial \overrightarrow{A} / \partial t$

Assuming Lorentz gauge $\partial \phi / \partial t + \nabla \cdot \overrightarrow{A} = 0$, one can show $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

Going back to the 4-force density, $f_\mu = F_{\mu\nu} J^\nu = F_{\mu\nu} \partial_\lambda F^{\nu\lambda}$ using $J^\mu = \partial_\nu F^{\mu\nu}$

After some fun steps, we have $f^\mu = \nabla_\nu (F^{\mu\lambda} F^\nu_{\ \lambda} - \frac{1}{4} g^{\mu\nu} F_{\lambda\kappa} F^{\lambda\kappa}) = \nabla_\nu T^{\mu\nu}_{\mathrm{EM}}$

where $T^{\mu\nu}_{\mathrm{EM}}$ is the electromagnetic stress-energy tensor

Stress-Energy Tensor $T^{\mu\nu}$

Physical interpretation for a fluid: flux of the μth component of the energy or momentum vector through a hyper-surface perpendicular to $\hat{e}_\nu$

$$\mu, \nu \equiv (t, \vec{x}) \qquad T^{\mu\nu} = \begin{pmatrix} \text{energy density} & \text{momentum density} \\ \text{momentum flux} & \text{stress tensor} \end{pmatrix}$$

Stress tensor:

$$T^{ii} = p^i \quad \text{(Pressure density vector)}$$

$$T^{ij} = \sigma^{ij} \text{ for } i \neq j \quad \text{(Shear stress density tensor)}$$

$$T^{\mu\nu}_{\text{MHD}} = T^{\mu\nu}_{\text{fluid}} + T^{\mu\nu}_{EM} = (\rho\epsilon u^\mu u^\nu + p g^{\mu\nu}) + (F^{\mu\lambda}F^\nu_\lambda - \frac{1}{4} g^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta})$$

$\epsilon = 1 + U + P/\rho$ Fluid part EM part

Stress-Energy Tensor $T^{\mu\nu}$

$$T^{\mu\nu}_{\rm MHD} = T^{\mu\nu}_{\rm fluid} + T^{\mu\nu}_{EM} = (\rho\epsilon u^\mu u^\nu + p g^{\mu\nu}) + (F^{\mu\gamma}F^{\nu}_{\gamma} - \frac{1}{4} g^{\mu\nu} F^{\alpha\beta}F_{\alpha\beta})$$

Poynting flux is the EM energy flux in the *i*-th (spatial) direction

$$T^{\mu}_{\nu} = F^{\mu\gamma}F_{\nu\gamma} - \frac{1}{4}(F_{\gamma\beta}F^{\gamma\beta})\delta^{\mu}_{\nu}$$

Therefore, outgoing radial Poynting flux: $S_r = -cT^{r}_{t,\rm EM} = -E \times B$

Poynting flux usually plays a dominant role in outflowing power as
jets and winds from compact objects are usually shown to be magnetically-driven

General relativistic MHD

Gammie+03,
Komissarov 99

Mass conservation $\nabla_\mu(\rho u^\mu) = 0,$

Expanding $\nabla_\mu, \Rightarrow \nabla_\mu(\rho u^\mu) = \frac{\partial(\rho u^\mu)}{\partial x^\mu} + \Gamma^\lambda_{\lambda\mu}\rho u^\mu = \frac{\partial}{\partial x^\mu}(\sqrt{-g}\rho u^\mu)$

$\Gamma^\mu_{\nu\lambda}$: Christoffel symbol and $\Gamma^\lambda_{\lambda\mu} = (1/2)g^{\delta\nu}\partial_\mu g_{\delta\nu} = \partial_\mu(\ln g) = (1/\sqrt{-g})\partial_\mu\sqrt{-g}$ where $g \equiv \mathrm{Det}(g_{\mu\nu})$

Energy-momentum conservation: $\nabla_\mu T^{\mu\nu} = 0$ (and is tied to the Einstein tensor $G^{\mu\nu} = 8\pi T^{\mu\nu}$
which works well since the covariant derivative of the Einstein tensor also vanishes)

The ideal Ohm's law becomes $u_\mu F^{\mu\nu} = 0$

And we have Maxwell's equations: $\nabla_\mu F^{\mu\nu} = J^\mu,\ \nabla_\mu {}^*F^{\mu\nu} = 0$

We can construct a 4-magnetic field: $b^\mu = (1/2)\epsilon^{\mu\nu\kappa\lambda}u_\nu F_{\kappa\lambda}$

b^μ is actually the co-moving magnetic field as $b^t = 0, b^i = B^i$ in the fluid rest frame

Also $b_\mu u^\mu = 0$, which gives us $F^{\mu\nu} = \epsilon^{\mu\nu\kappa\lambda}u_\kappa b_\lambda$ and ${}^*F^{\mu\nu} = b^\mu u^\nu - b^\nu u^\mu$

Finally, $T^{\mu\nu}_{\mathrm{EM}} = b^2 u^\mu u^\nu + (b^2/2)g^{\mu\nu} - b^\mu b^\nu$, where $b^2 = B^2 - E^2$

Black Hole spacetime metrics

Horizon penetrating form: Kerr-Schild Coordinates

McKinney & Gammie 04

The transformation matrix from BL to KS is

$$\frac{\partial t[\mathrm{KS}]}{\partial r[\mathrm{BL}]} = \frac{2r}{\Delta},$$

and

$$\frac{\partial \phi[\mathrm{KS}]}{\partial r[\mathrm{BL}]} = \frac{a}{\Delta};$$

$$\begin{aligned} ds^2 = &- \left(1 - \frac{2r}{\Sigma}\right) dt^2 + \left(\frac{4r}{\Sigma}\right) dr\, dt + \left(1 + \frac{2r}{\Sigma}\right) dr^2 + \Sigma\, d\theta^2 \\ &+ \sin^2\theta \left[\Sigma + a^2 \left(1 + \frac{2r}{\Sigma}\right) \sin^2\theta\right] d\phi^2 \\ &- \left(\frac{4ar \sin^2\theta}{\Sigma}\right) d\phi\, dt - 2a \left(1 + \frac{2r}{\Sigma}\right) \sin^2\theta\, d\phi\, dr, \end{aligned}$$

Removes the coordinate singularity seen in the BL coordinate metric form

Widely used in numerical simulations

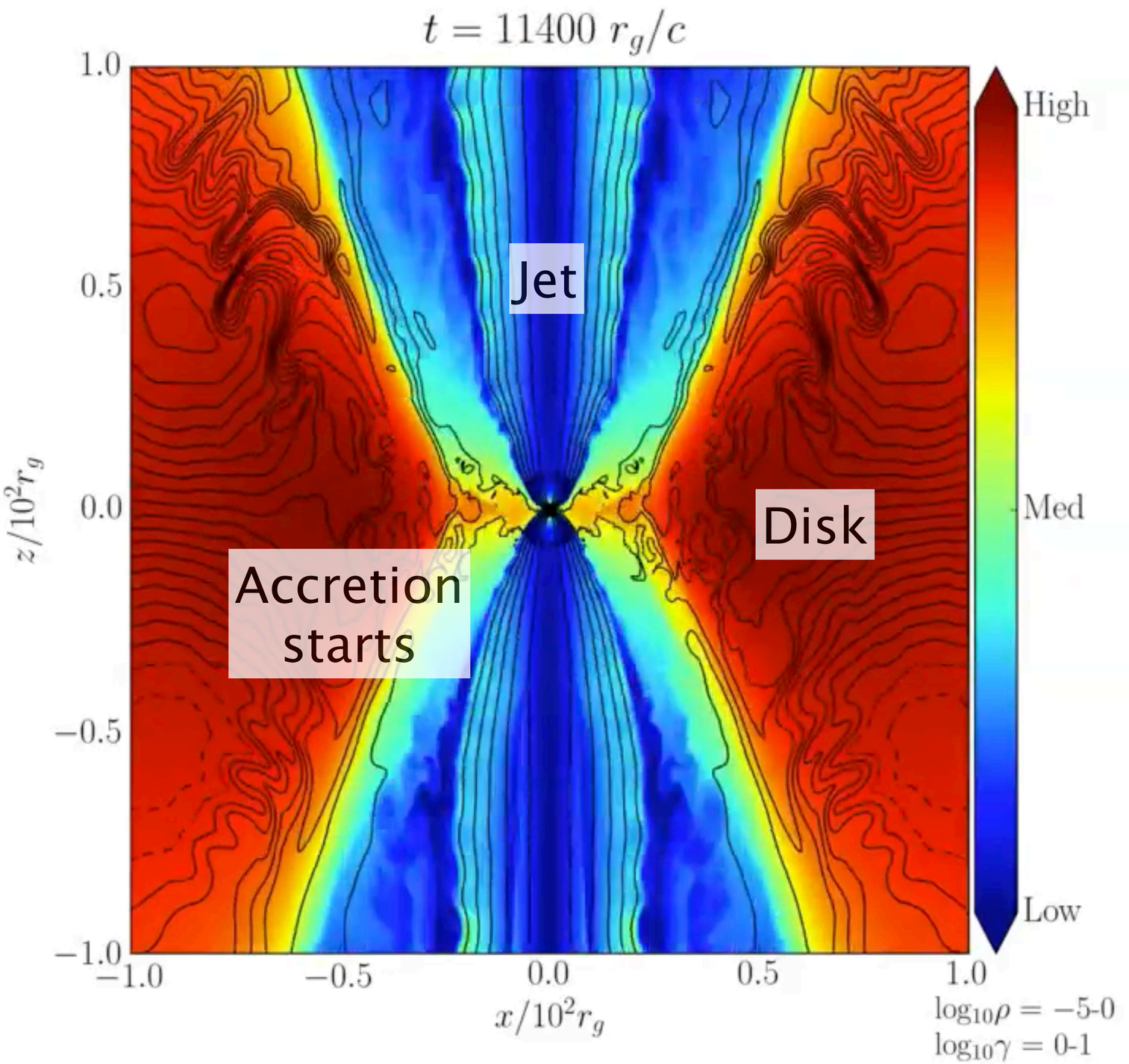
3+1 decomposition is a popular way of rewriting the GRMHD equations (Valencia formulation)

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt)$$

General relativistic MHD simulations

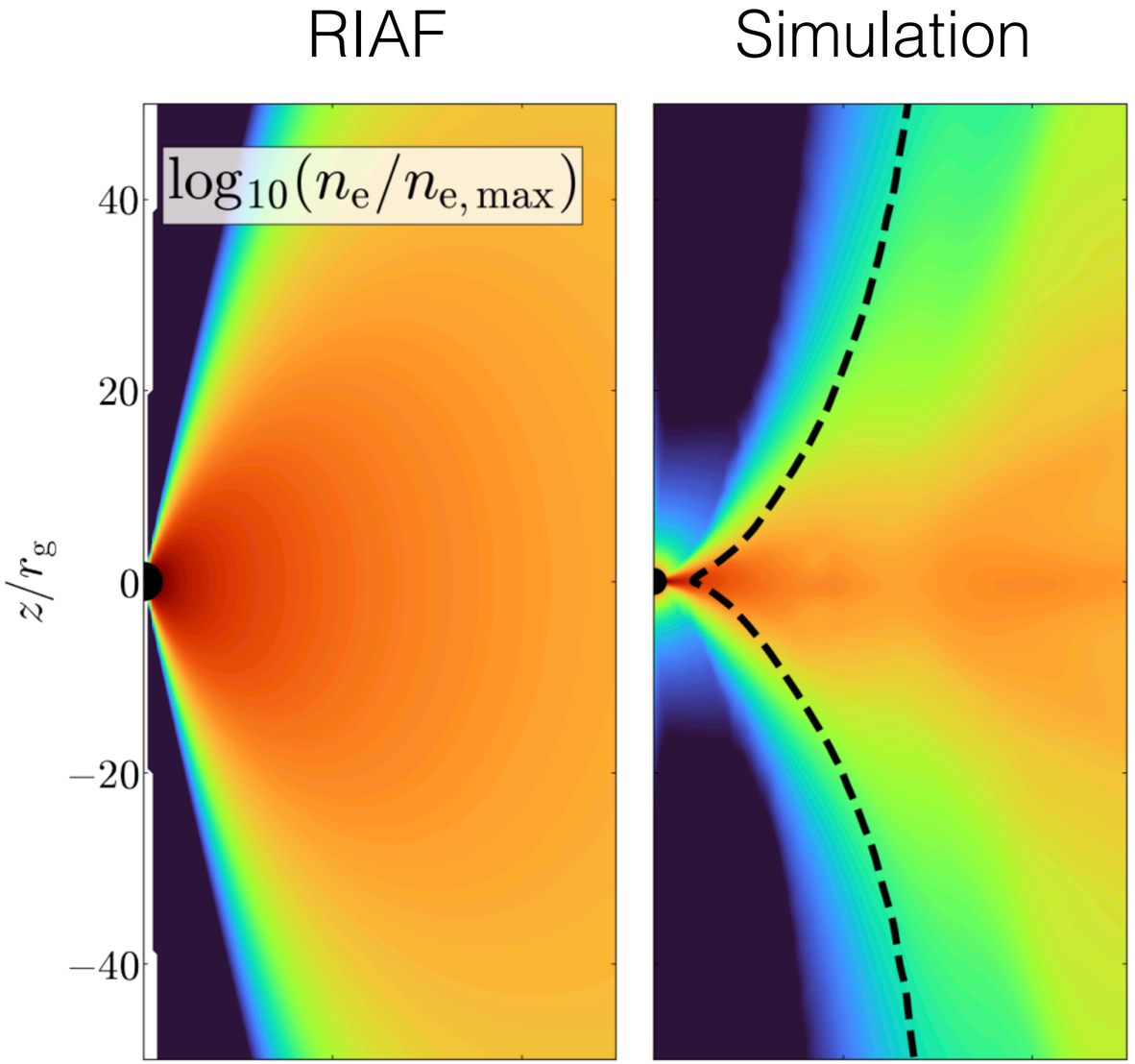
- Turbulent, magnetized disk
- Highly magnetized, fast, near-vacuum outflow (jet)
- Slow, gas-rich disk outflow (wind)

A movie of the
fluid density (High->disk)
and
Lorentz factor (High->jet)



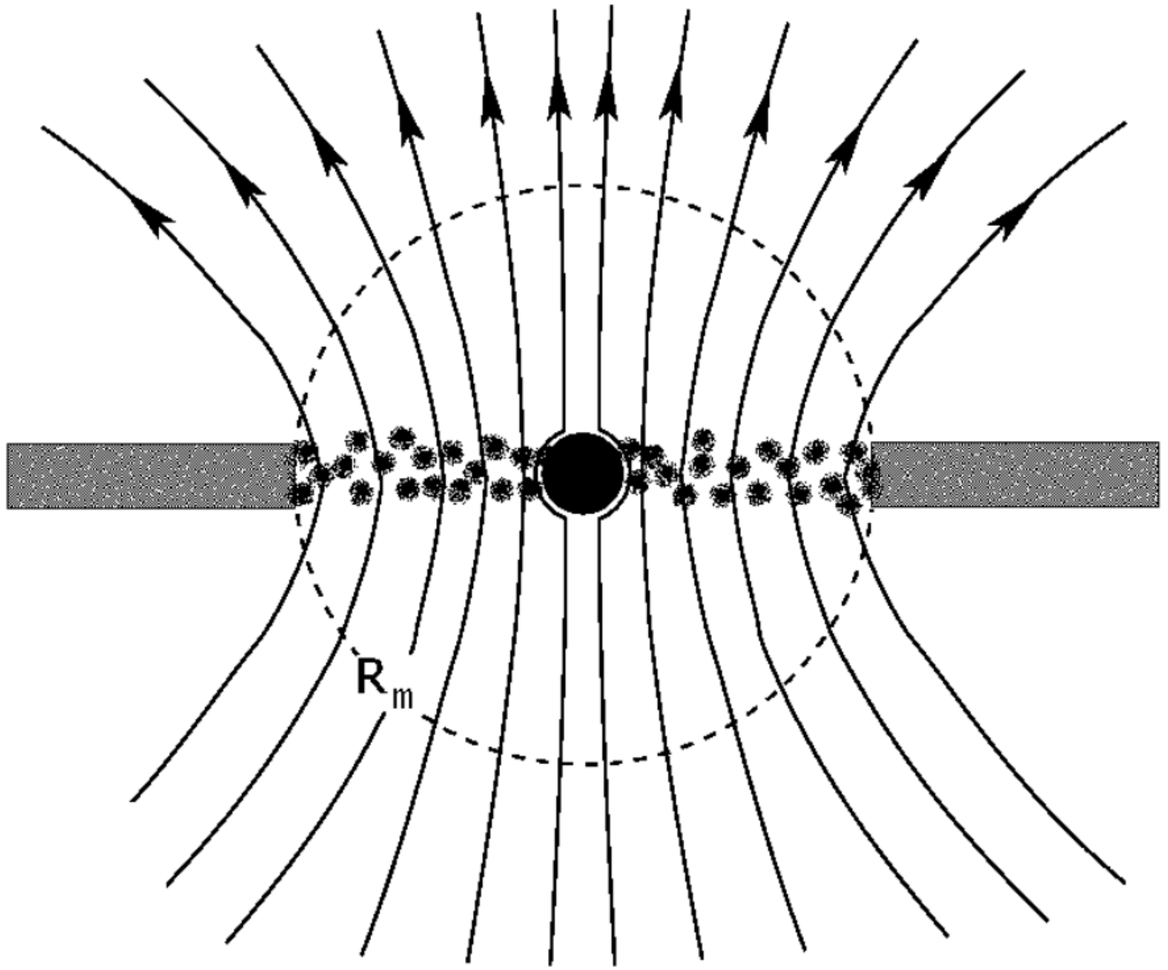
Advective inflows

- Semi-analytical disk solutions are excellent guides for building intuition about the average properties of the disk
- Different geometrical shape
- Presence of outflows such as jets
- Effect of magnetic fields



Advective inflows

- Semi-analytical disk solutions are excellent guides for building intuition about the average properties of the disk
- Different geometrical shape
- Presence of outflows such as jets
- Effect of magnetic fields



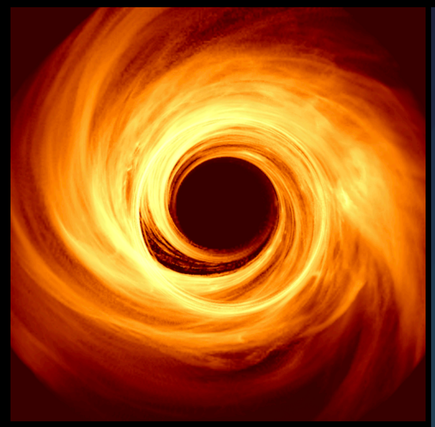
Narayan+03

Horizon structure in accreting BHs

- Gas accretion onto black holes bring along magnetic fields
- Huge influx of magnetic fields -> Magnetically Arrested Disk (MAD)
- Over-saturation drives magnetic eruptions

Igumenshchev+03, Narayan+03, Tchekhovskoy+11, Begelman++, Chael++, Chatterjee++, Davelaar++, Dexter++, Liska++, McKinney++, Porth++, Ressler++, Ricarte++, Ripperda++, White++ and many more

Quiescent

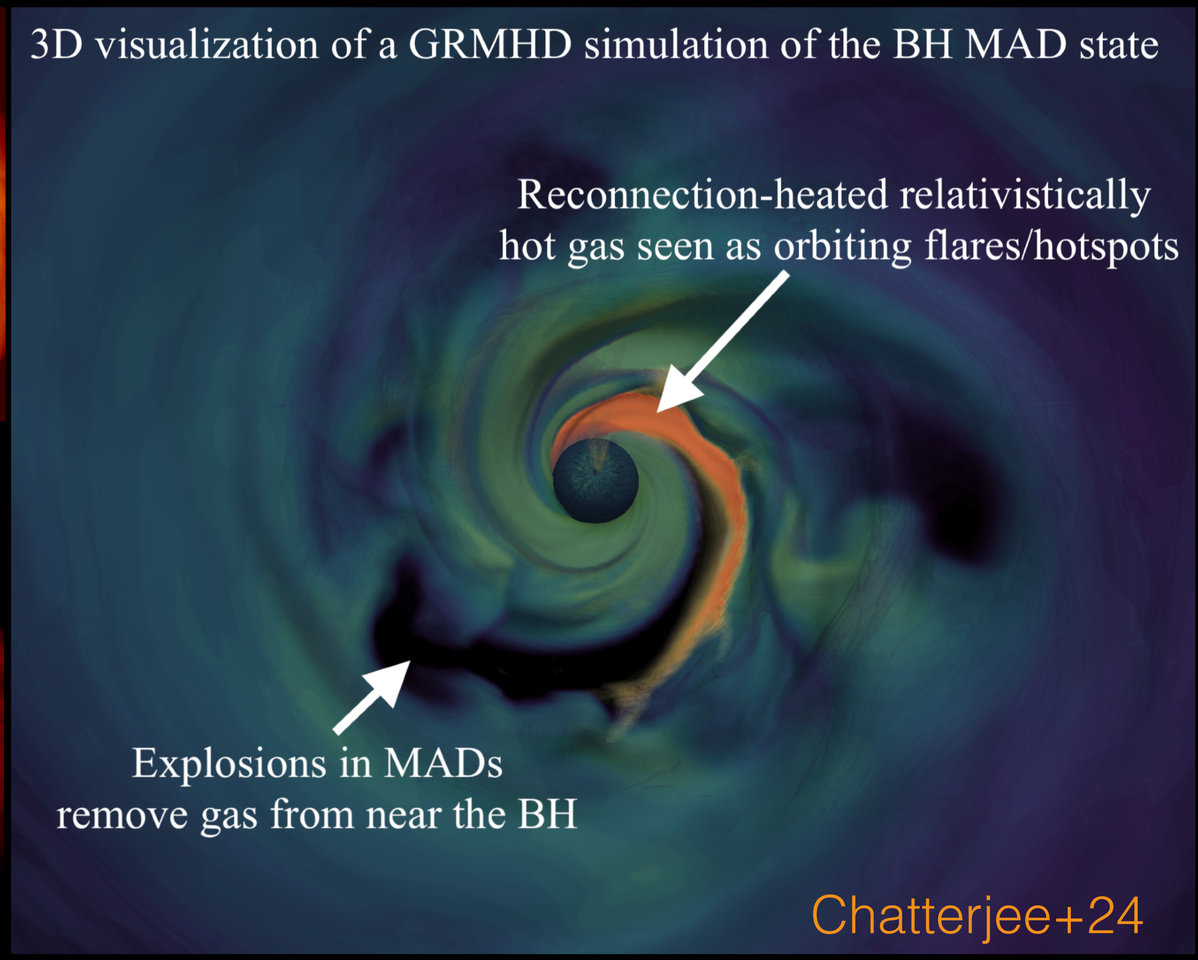


Flaring



Gelles+22

MADs in action

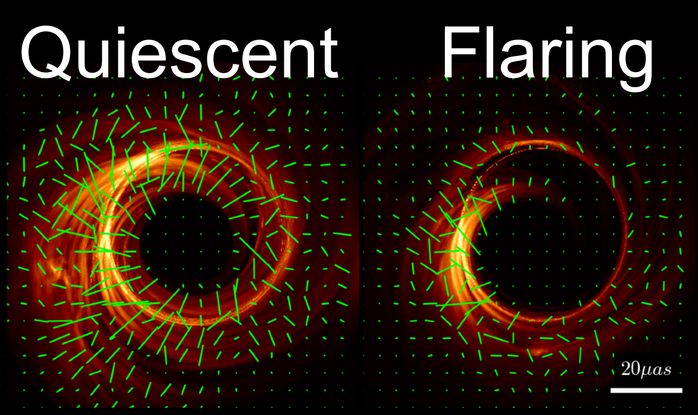


Chatterjee+24

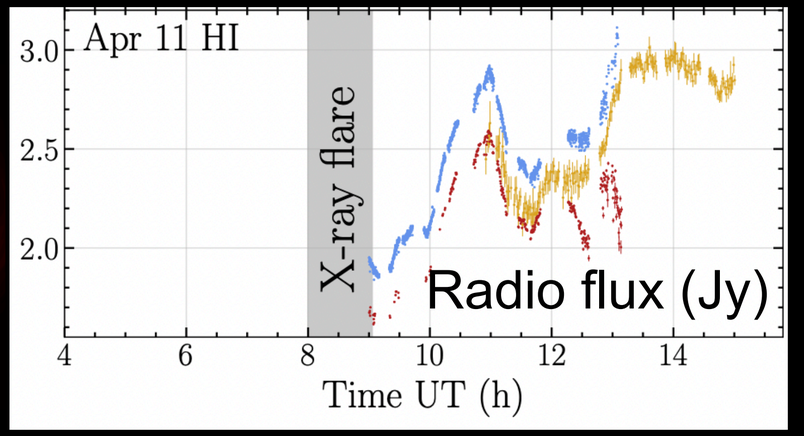
Flares, thin disks, truncation and more…

Detailed Horizon-scale Activity

- Polarized imaging of Black Holes
- Tracking high-energy emission in accretion flows and jets

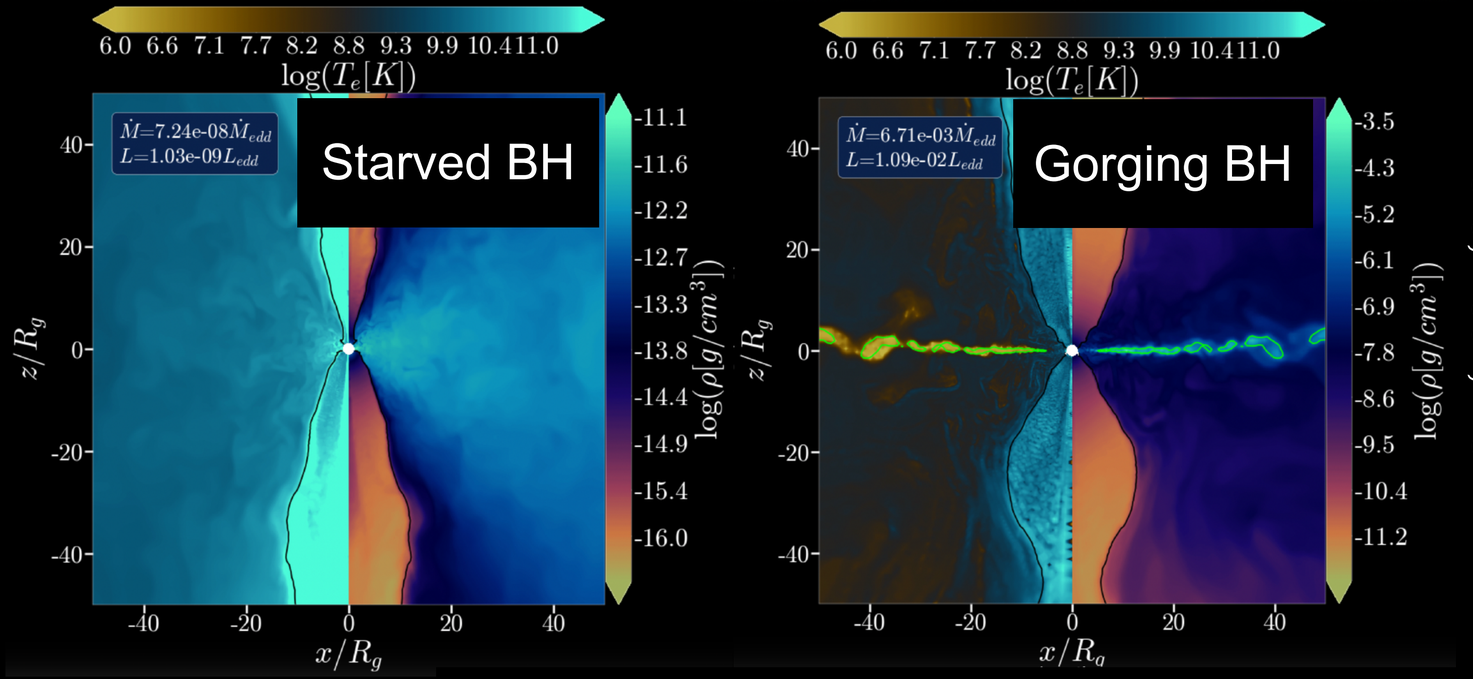


Wielgus, EHTC+22 (incl. KC)



Radiation physics

- Rapidly accreting BHs emit a lot of photons: radiative feedback
- Much more numerically expensive



Liska, Kaaz, KC+24